

# TarangNet™ API Documentation

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## For TarangMini™ STxx

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## Contents

|                                                       |    |
|-------------------------------------------------------|----|
| 1. TarangMini™ .....                                  | 3  |
| 2. TarangNet™ Stack for TarangMini™ .....             | 3  |
| 5. API Interface .....                                | 4  |
| 5.1. Invalid Command Response .....                   | 5  |
| 5.2. Module Configuration Commands .....              | 5  |
| 5.2.1. List of Supported Configuration Commands ..... | 6  |
| 5.3. Module Communication Commands .....              | 7  |
| 5.4. Unsolicited Event Alerts .....                   | 7  |
| 6. Configuration Commands Description .....           | 8  |
| 6.1. Restore Defaults (W) .....                       | 8  |
| 6.2. Restart Module (W) .....                         | 9  |
| 6.3. Flash Write (W) .....                            | 9  |
| 6.4. Serial Baud rate (R/W) .....                     | 10 |
| 6.5. Device EUI Read (R) .....                        | 11 |
| 6.6. Module Hardware Version Read (R) .....           | 11 |
| 6.7. Firmware Version Read (R) .....                  | 12 |
| 6.8. Confirm Message (R/W) .....                      | 12 |
| 6.9. Default Data Rate (R/W) .....                    | 13 |
| 6.10. Sleep Time (R/W) .....                          | 15 |
| 6.11. Wake Time (R/W) .....                           | 16 |
| 6.12. Sleep Mode (R/W) .....                          | 17 |
| 6.13. Max Output Power (R/W) .....                    | 18 |
| 6.14. RSSI and SNR Read (R) .....                     | 19 |
| 6.15. Default Frequency (R/W) .....                   | 19 |
| 6.16. Destination Address (R/W) .....                 | 20 |
| 6.17. Extended Payload Information (R/W) .....        | 21 |
| 6.18. Receive Address (R) .....                       | 22 |
| 6.19. Pan Id (R/W) .....                              | 23 |
| 6.20. Multicast Address (R/W) .....                   | 24 |
| 6.21. Connection Type(R/W) .....                      | 25 |
| 6.22. Mode selection (R/W) .....                      | 26 |

## 1. TarangMini™

Melange Systems TarangMini™ family of RF modules provide a common footprint which can host multiple platforms, including TarangNet™ (Point to Point, Point to Multi-point), Pretty mesh™ (for mesh network), 6LoWPAN and LoRaWAN™. These modules are designed to address the unique needs of low- power, low-cost wireless sensor and control networks. TarangMini™ modules support both AT and API interface to the host. Both have their own unique use cases.

## 2. TarangNet™ Stack for TarangMini™

TarangNet™ defines the communication protocol and system architecture for the network while the LoRa® physical layer enables the long-range communication link. The protocol and network architecture have the most influence in determining the battery lifetime of a node, the network capacity, the quality of service, the security, and the variety of applications served by the network. TarangNet™ enables users to communicate with multiple TarangMini™ RF modules without additional hardware requirement.

## 3. TarangNet™: Pin Definition

| Pin | Name      | Type               | Function                                           |
|-----|-----------|--------------------|----------------------------------------------------|
| 1   | VDD       | Power Supply       | Supply Voltage                                     |
| 2   | Module_TX | Output [Push-pull] | Serial data OUT from TarangMini™                   |
| 3   | Module_RX | Input              | Serial data IN to TarangMini™                      |
| 4   | BOOT      | Input [Pulled-up]  | High - Flash Programming<br>Low - Normal Operation |
| 5   | RESET     | Input [Pulled-up]  | Reset module (Active Low)                          |
| 6   | GND       | Power Supply       | Ground                                             |
| 7   | NC        | Unused (tri-state) | NA                                                 |
| 8   | NC        | Unused (tri-state) | NA                                                 |
| 9   | NC        | Unused (tri-state) | NA                                                 |
| 10  | NC        | Unused (tri-state) | NA                                                 |
| 11  | NC        | Unused (tri-state) | NA                                                 |
| 12  | NC        | Unused (tri-state) | NA                                                 |
| 13  | SLEEP     | Input              | High – Normal Mode<br>Low – Sleep Mode             |
| 14  | NC        | Unused (tri-state) | NA                                                 |
| 15  | NC        | Unused (tri-state) | NA                                                 |
| 16  | NC        | Unused (tri-state) | NA                                                 |
| 17  | NC        | Unused (tri-state) | NA                                                 |
| 18  | MODE      | Input              | High - API Window<br>Low - Transparent Window      |

## 4. UART Modes

TarangMini™ module has a standard UART interface to connect with an external controller or a PC. The module treats the UART data based on the current status of MODE Selection API. It's controlled base on Mode selection API Command and Pin status.

This module supports two working modes:

- Transparent Window
- API Window

In "Transparent Window mode any data sent to the module UART is treated as payload. A TarangMini™ module wirelessly transmits this UART data to the other TarangMini™ modules in range.

Asserting the "MODE" input pin to logic "High" makes the module to open an "API Window" on UART. Here the data sent on the UART is treated as an API Command, checked for its correctness and executed as per the command function. It is also possible to send and receive the wireless data during the "API Window" except during the execution of few commands that involves device configuration changes or non-volatile parameter writes. Refer [\(Mode selection \(R/W\)\)](#).

## 5. API Interface

API mode is a frame-based protocol that allows to direct data on a packet basis. Data packets are created with needed information, such as Start Byte, Packet Length, Packet Type, Command ID and Payload.

Every UART API issued by the host will be acknowledged by the Module. Host is recommended to wait minimum 500 mSec for any response before initiating a timeout action. Module reset or power-cycle are the recommended timeout actions.

Every UART API must be issued as a single packet. There should not be any inter-character delay within a packet. Multiple APIs cannot be clubbed together, module will serve only single API at any instance, host must wait and ensure a valid response for each APIs before issuing a new one.

### Generic Command Packet Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte)             | Packet Type<br>(1 byte)                                                                                                                     | Packet Specific Structure |
|------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 0x2B                   | Packet Length<br>excluding Start Byte | 0x02: Configuration Command<br>0x04: Router mode Communication<br>Command<br>0x05: Root mode communication<br>Command + Destination Address |                           |

### Generic Response Packet Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte)             | Packet Type<br>(1 byte)                                                                                                                                                                                           | Packet Specific<br>Structure |
|------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 0x2B                   | Packet Length<br>excluding Start Byte | 0x05: Module Startup Response<br>0x80: Unicast Receive Response<br>0x81: Sent Data Response<br>0x82: Configuration Response<br>0x83: Multicast Receive Response<br>0x84: Sleep Response<br>0x88: Invalid Response |                              |

#### 5.1. Invalid Command Response

An Invalid Command Response is issued if the module receives any UART data other than the specified structures. Following packet shall be pushed as a response.

### Invalid Command Response Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Payload |
|------------------------|---------------------------|-------------------------|---------|
| 0x2B                   | 0x03                      | 0x88                    | 0x01    |

#### 5.2. Module Configuration Commands

Module configuration commands are used for reading and writing the module parameters. A flash write API to be issued at the end, if parameters configured by other APIs to be retained after a power cycle or module reset.

Some of the parameters mandatorily requires a flash write and a reset to take effect, please refer individual API description. A restart by software also can be performed using an API after issuing flash write API.

### Configuration Command Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte)                | Packet Type<br>(1 byte) | Read/Write<br>(1 byte) | Command ID<br>(2 byte) | Value /<br>Payload<br>(Optional) |
|------------------------|------------------------------------------|-------------------------|------------------------|------------------------|----------------------------------|
| 0x2B                   | Packet Length<br>excluding Start<br>Byte | 0x02                    | 00 Read<br>01 Write    |                        |                                  |

### Configuration Command Response Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte)                | Packet Type<br>(1 byte) | Command ID<br>(2 byte) | Execution Status<br>(1 byte)                                | Value<br>(Optional) |
|------------------------|------------------------------------------|-------------------------|------------------------|-------------------------------------------------------------|---------------------|
| 0x2B                   | Packet Length<br>excluding Start<br>Byte | 0x82                    |                        | 00 Success<br>01 Invalid Parameter<br>02 Invalid Command ID |                     |

#### 5.2.1. List of Supported Configuration Commands

| Sl.No | API Command                  | Command ID | R/W |
|-------|------------------------------|------------|-----|
| 1     | Restore Defaults             | 00 00      | W   |
| 2     | Restart Module               | 00 01      | W   |
| 3     | Flash Write                  | 00 02      | W   |
| 4     | Serial Baudrate              | 00 03      | R/W |
| 5     | Device EUI Read              | 00 04      | R   |
| 6     | Module Hardware Version Read | 00 05      | R   |
| 7     | Firmware Version Read        | 00 06      | R   |
| 8     | Confirm Message              | 00 07      | R/W |
| 9     | Default Data Rate            | 00 08      | R/W |
| 10    | Sleep Time                   | 00 09      | R/W |
| 11    | Wake Time                    | 00 0A      | R/W |
| 12    | Sleep Mode                   | 00 0B      | R/W |
| 13    | Max Output Power             | 00 0C      | R/W |
| 14    | RSSI and SNR Read            | 00 0D      | R   |
| 15    | Default Frequency            | 00 0F      | R/W |
| 16    | Destination Address          | 00 10      | R/W |
| 17    | Extended Payload Information | 00 11      | R   |
| 18    | Receive Address              | 00 12      | R   |
| 19    | PAN ID                       | 00 13      | R/W |
| 20    | Multicast Address            | 00 14      | R/W |
| 21    | Connection Type              | 00 15      | R/W |
| 22    | Mode selection               | 00 16      | R/W |

### 5.3. Module Communication Commands

#### Data Uplink Command & Response Structure

| Communication Type            | API Packet Structure   |                           |                         |                                       |
|-------------------------------|------------------------|---------------------------|-------------------------|---------------------------------------|
|                               | Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Payload                               |
| Router mode Data Send Command | 0x2B                   | 2 + Payload Bytes         | 0x04                    | <Payload>                             |
| Root mode Data Send Command   | 0x2B                   | 2 + 8 + Payload Bytes     | 0x05                    | <Destination address> + <Payload>     |
| Data Send Response            | 0x2B                   | 0x03                      | 0x81                    | 0x00: Uplink ACK<br>0x01: Uplink NACK |

### 5.4. Unsolicited Event Alerts

Module supports unsolicited event alert feature to ease the host application integration. This feature will result in getting unsolicited notifications for Module Startup Event, Module Join Event, Module Sleep Event and Downlink Data Reception Events.

#### Unicast Receive Data Response Structure

| Communication Type                                  | API Packet Structure   |                           |                         |                                     |                             |                  |                 |           |  |
|-----------------------------------------------------|------------------------|---------------------------|-------------------------|-------------------------------------|-----------------------------|------------------|-----------------|-----------|--|
|                                                     | Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Additional Payload Info<br>(1 byte) | Payload                     |                  |                 |           |  |
| Unicast Receive                                     | 0x2D                   | 3+Payload Bytes           | 0x80                    | 0x00                                | <Payload>                   |                  |                 |           |  |
| Unicast Receive with Additional Payload Information | 0x2D                   | 15+Payload Bytes          | 0x80                    | 0x01                                | <Receive Address> (8 bytes) | <RSSI> (2 bytes) | <SNR> (2 bytes) | <Payload> |  |

#### Multicast Receive Data Response Structure

| Communication Type                                    | API Packet Structure   |                           |                         |                                     |                             |                  |                 |           |  |
|-------------------------------------------------------|------------------------|---------------------------|-------------------------|-------------------------------------|-----------------------------|------------------|-----------------|-----------|--|
|                                                       | Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Additional Payload Info<br>(1 byte) | Payload                     |                  |                 |           |  |
| Multicast Receive                                     | 0x2D                   | 3+Payload Bytes           | 0x83                    | 0x00                                | <Payload>                   |                  |                 |           |  |
| Multicast Receive with Additional Payload Information | 0x2D                   | 15+Payload Bytes          | 0x83                    | 0x01                                | <Receive Address> (8 bytes) | <RSSI> (2 bytes) | <SNR> (2 bytes) | <Payload> |  |

### Module Start-up Serial Response Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Payload                                        |
|------------------------|---------------------------|-------------------------|------------------------------------------------|
| 0x2B                   | 0x11                      | 0x05                    | 0x54 61 72 61 6E 67 4E 65 74 5F 53 74 61 72 74 |

### Module Sleep Response Structure

| Start Byte<br>(1 byte) | Packet Length<br>(1 byte) | Packet Type<br>(1 byte) | Payload                                                   |
|------------------------|---------------------------|-------------------------|-----------------------------------------------------------|
| 0x2B                   | 0x03                      | 0x84                    | 0x00: Module in Sleep Mode<br>0x01: Module in Normal Mode |

## 6. Configuration Commands Description

### 6.1. Restore Defaults (W)

This command restores module configuration to factory default settings

| Command Structure |               |             |             |            |      |
|-------------------|---------------|-------------|-------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Write | 0x00       | 0x00 |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x00 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

<\*> A module restart is required for this API operation to take effect

<\*> This API will take up to 2 seconds to issue response

## 6.2. Restart Module (W)

This command restarts the module

| Command Structure |               |             |             |            |      |
|-------------------|---------------|-------------|-------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Write | 0x00       | 0x01 |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x01 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

## 6.3. Flash Write (W)

This command saves the module configuration to flash

| Command Structure |               |             |             |            |      |
|-------------------|---------------|-------------|-------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Write | 0x00       | 0x02 |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x02 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

<\*> This API will take up to 2 seconds to issue response

## 6.4. Serial Baud rate (R/W)

This command enables the user to set or read the module serial Baud rate.

| Command Structure |               |             |             |            |      |                                                                                                                                                             |
|-------------------|---------------|-------------|-------------|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                                                                                                                                          |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x03 | 0x00: 1200 bps<br>0x01: 2400 bps<br>0x02: 4800 bps<br>0x03: 9600 bps (Default)<br>0x04: 19200 bps<br>0x05: 38400 bps<br>0x06: 57600 bps<br>0x07: 115200 bps |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x03 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read Operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x03 |

| Response Structure |               |             |            |      |                                                                  |                                                                                                                                                             |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)                                                                                                                                          |
| 0x2B               | 0x06          | 0x02        | 0x00       | 0x03 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: 1200 bps<br>0x01: 2400 bps<br>0x02: 4800 bps<br>0x03: 9600 bps (Default)<br>0x04: 19200 bps<br>0x05: 38400 bps<br>0x06: 57600 bps<br>0x07: 115200 bps |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.5. Device EUI Read (R)

This command reads the module Device EUI.

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x04 |

| Response Structure |               |             |            |      |                                                                  |                       |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-----------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 8 Bytes) |
| 0x2B               | 0x0D          | 0x82        | 0x00       | 0x04 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | < Device EUI >        |

## 6.6. Module Hardware Version Read (R)

This command reads the module Hardware Version.

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x05 |

| Response Structure |               |             |            |      |                                                                  |                      |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|----------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (ASCII)        |
| 0x2B               | <length>      | 0x82        | 0x00       | 0x05 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | < Hardware Version > |

## 6.7. Firmware Version Read (R)

This command reads the module Firmware version.

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x06 |

| Response Structure |               |             |            |      |                                                                  |                      |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|----------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (ASCII)        |
| 0x2B               | <length>      | 0x82        | 0x00       | 0x06 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | < Firmware Version > |

## 6.8. Confirm Message (R/W)

This command enables the user to set or read Confirm Message parameter.

| Command Structure |               |             |             |            |      |                                         |
|-------------------|---------------|-------------|-------------|------------|------|-----------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (HEX - 1 Byte)                    |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x07 | 0x00: Disable (Default)<br>0x01: Enable |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x07 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x07 |

| Response Structure |               |             |            |      |                                                                  |                                        |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|----------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 1 Byte)                   |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x07 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: Disable(Default)<br>0x01: Enable |

## 6.9. Default Data Rate (R/W)

This command enables the user to set or read the Default Data Rate parameter.

| Command Structure |               |             |             |            |      |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|---------------|-------------|-------------|------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                                                                                                                                                                                                                                                                                                                                                                               |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x08 | 0x00: SF12, BW125 (Default)<br>0x01: SF12, BW250<br>0x02: SF12, BW250<br>0x03: SF11, BW125<br>0x04: SF11, BW250<br>0x05: SF11, BW500<br>0x06: SF10, BW125<br>0x07: SF10, BW250<br>0x08: SF10, BW500<br>0x09: SF09, BW125<br>0x0A: SF09, BW250<br>0x0B: SF09, BW500<br>0x0C: SF08, BW125<br>0x0D: SF08, BW250<br>0x0E: SF08, BW500<br>0x0F: SF07, BW125<br>0x10: SF07, BW250<br>0x11: SF07, BW500 |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x08 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x08 |

| Response Structure |               |             |            |      |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x08 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: SF12, BW125 (Default)<br>0x01: SF12, BW250<br>0x02: SF12, BW250<br>0x03: SF11, BW125<br>0x04: SF11, BW250<br>0x05: SF11, BW500<br>0x06: SF10, BW125<br>0x07: SF10, BW250<br>0x08: SF10, BW500<br>0x09: SF09, BW125<br>0x0A: SF09, BW250<br>0x0B: SF09, BW500<br>0x0C: SF08, BW125<br>0x0D: SF08, BW250<br>0x0E: SF08, BW500<br>0x0F: SF07, BW125<br>0x10: SF07, BW250<br>0x11: SF07, BW500<br>0x12: SF06, BW125<br>0x13: SF06, BW250<br>0x14: SF06, BW500<br>0x15: SF05, BW125<br>0x16: SF05, BW250<br>0x17: SF05, BW500 |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.10. Sleep Time (R/W)

This command enables the user to set or read the Sleep Time parameter (in Seconds).

| Command Structure |               |             |             |            |      |                                 |
|-------------------|---------------|-------------|-------------|------------|------|---------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (HEX - 2 Byte)            |
| 0x2B              | 0x07          | 0x02        | 0x01: Write | 0x00       | 0x09 | <Sleep Time> (0x00 14: Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x09 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x09 |

| Response Structure |               |             |            |      |                                                                  |                                 |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|---------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 2 Byte)            |
| 0x2B               | 0x07          | 0x82        | 0x00       | 0x09 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Sleep Time> (0x00 14: Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

### 6.11. Wake Time (R/W)

This command enables the user to set or read Wake Time parameter (in Seconds).

| Command Structure |               |             |             |            |      |                                |
|-------------------|---------------|-------------|-------------|------------|------|--------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (HEX - 2 Byte)           |
| 0x2B              | 0x07          | 0x02        | 0x01: Write | 0x00       | 0x0A | <Wake Time> (0x00 14: Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x0A | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |
|-------------------|---------------|-------------|------------|------------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00 0x0A  |

| Response Structure |               |             |            |      |                                                                  |                                 |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|---------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 2 Byte)            |
| 0x2B               | 0x07          | 0x82        | 0x00       | 0x0A | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Sleep Time> (0x00 14: Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.12. Sleep Mode (R/W)

This command enables the user to set or read the Sleep Mode configuration.

| Command Structure |               |             |             |            |      |                                                                                     |
|-------------------|---------------|-------------|-------------|------------|------|-------------------------------------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                                                                  |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x0B | 0x00: No Sleep <b>(Default)</b><br>0x01: Pin Asserted Sleep<br>0x02: Periodic Sleep |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x0B | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x0B |

| Response Structure |               |             |            |      |                                                                  |                                                                                     |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)                                                                  |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x0B | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: No Sleep <b>(Default)</b><br>0x01: Pin Asserted Sleep<br>0x02: Periodic Sleep |

<\*> A flash write and module restart is required for this API write operation to take effect

### 6.13. Max Output Power (R/W)

This command enables the user to set or read Max Output Power of the module. Please refer the hardware documentation to map these values to corresponding power index.

| Command Structure |               |             |             |            |      |                                   |
|-------------------|---------------|-------------|-------------|------------|------|-----------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x0C | <Max Output Power>(0x16: Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x0C | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x0C |

| Response Structure |               |             |            |      |                                                                  |                                   |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-----------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)                |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x0C | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Max Output Power>(0x16: Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

#### 6.14. RSSI and SNR Read (R)

This command reads the RSSI and SNR values of the last received packet (in ASCII format).

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x0D |

| Response Structure |               |             |            |      |                                                                  |                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (ASCII )   |
| 0x2B               | <Length>      | 0x82        | 0x00       | 0x0D | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | < RSSI and SNR > |

#### 6.15. Default Frequency (R/W)

This command allows the user to read or write transmission/reception frequency.

| Command Structure |               |             |             |            |      |                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|---------------|-------------|-------------|------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-4 Byte)                                                                                                                                                                                                                                                                                                                                           |
| 0x2B              | 0x09          | 0x02        | 0x01: Write | 0x00       | 0x0F | Set value<br>339060E0: 865.1MHz(Default)<br>33936E20: 865.3MHz<br>33967B60: 865.5MHz<br>339988A0: 865.7MHz<br>339C95E0: 865.9MHz<br>339FA320: 866.1MHz<br>33A2B060: 866.3MHz<br>33A5BDA0: 866.5MHz<br>33A8CAE0: 866.7MHz<br>33ABD820: 866.9MHz<br>33AEE560: 867.1MHz<br>33B1F2A0: 867.3MHz<br>33B4FFE0: 867.5MHz<br>33B80D20: 867.7MHz<br>33BB1A60: 867.9MHz |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x0F | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

## Read Operation

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x0F |

| Response Structure |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------|---------------|-------------|------------|------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status    | Value (Hex-4 Byte)                                                                                                                                                                                                                                                                                                                                     |
| 0x2B               | 0x09          | 0x82        | 0x00       | 0x0F | 0x00: Success       | <b>339060E0: 865.1MHz(Default)</b><br>33936E20: 865.3MHz<br>33967B60: 865.5MHz<br>339988A0: 865.7MHz<br>339C95E0: 865.9MHz<br>339FA320: 866.1MHz<br>33A2B060: 866.3MHz<br>33A5BDA0: 866.5MHz<br>33A8CAE0: 866.7MHz<br>33ABD820: 866.9MHz<br>33AEE560: 867.1MHz<br>33B1F2A0: 867.3MHz<br>33B4FFE0: 867.5MHz<br>33B80D20: 867.7MHz<br>33BB1A60: 867.9MHz |
|                    |               |             |            |      | 0x01: Invalid value |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      | 0x02: Invalid       |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      | Command ID          |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |
|                    |               |             |            |      |                     |                                                                                                                                                                                                                                                                                                                                                        |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.16. Destination Address (R/W)

This command allows the user to read or write destination address to which module has to transmit data.

| Command Structure |               |             |             |            |      |                                                        |
|-------------------|---------------|-------------|-------------|------------|------|--------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-8 Byte)                                     |
| 0x2B              | 0x0D          | 0x02        | 0x01: Write | 0x00       | 0x10 | <Destination Address><br>(0xFFFFFFFFFFFFFFFF: Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x10 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

## Read Operation

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x10 |

| Response Structure |               |             |            |      |                                                                  |                                                        |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|--------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-8 Byte)                                     |
| 0x2B               | 0x0D          | 0x82        | 0x00       | 0x10 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Destination Address><br>(0xFFFFFFFFFFFFFFFF: Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.17. Extended Payload Information (R/W)

This command enables the user to set or read the Extended Payload Information, when this API is enabled then additional information such as RSSI, SNR and receive address will be concatenated with receive payload.

| Command Structure |               |             |             |            |      |                         |
|-------------------|---------------|-------------|-------------|------------|------|-------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)      |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x11 | 0x00: (Default)<br>0x01 |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x11 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

### Read Operation

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x11 |

| Response Structure |               |             |            |      |                                                                  |                         |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)      |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x11 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: (Default)<br>0x01 |

<\*> A flash write and module restart is required for this API write operation to take effect

### 6.18. Receive Address (R)

This command allows the user to read the receive address of the last received payload.

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x12 |

| Response Structure |               |             |            |      |                                                                  |                    |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|--------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-8 Byte) |
| 0x2B               | 0x0D          | 0x82        | 0x00       | 0x12 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Receive Address>  |

## 6.19. Pan Id (R/W)

This command enables the user to set or read the Pan Id of the module

| Command Structure |               |             |             |            |      |                               |
|-------------------|---------------|-------------|-------------|------------|------|-------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-2 Byte)            |
| 0x2B              | 0x07          | 0x02        | 0x01: Write | 0x00       | 0x13 | <Pan Id><br>(0xFFFF: Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x13 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x13 |

| Response Structure |               |             |            |      |                                                                  |                               |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-1 Byte)            |
| 0x2B               | 0x07          | 0x82        | 0x00       | 0x13 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Pan Id><br>(0xFFFF: Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.20. Multicast Address (R/W)

This command allows the user to read or write multicast address to which module has to transmit data.

| Command Structure |               |             |             |            |      |                                                         |
|-------------------|---------------|-------------|-------------|------------|------|---------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-8 Byte)                                      |
| 0x2B              | 0x0D          | 0x02        | 0x01: Write | 0x00       | 0x14 | <Multicast Address><br>(0x1122334455667788:<br>Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x14 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x14 |

| Response Structure |               |             |            |      |                                                                  |                                                         |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|---------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (Hex-8 Byte)                                      |
| 0x2B               | 0x0D          | 0x82        | 0x00       | 0x14 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | <Multicast Address><br>(0x1122334455667788:<br>Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.21. Connection Type(R/W)

This command allows the user to read or write module as root or router.

| Command Structure |               |             |             |            |      |                                      |
|-------------------|---------------|-------------|-------------|------------|------|--------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                   |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x15 | 0x00: Router (Default)<br>0x01: Root |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x15 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x15 |

| Response Structure |               |             |            |      |                                                                  |                                      |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|--------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 1 Byte)                 |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x15 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: Router (Default)<br>0x01: Root |

<\*> A flash write and module restart is required for this API write operation to take effect

## 6.22. Mode selection (R/W)

This command enables the user to configure the module for either API mode or Pin selection Mode. In Pin selection mode, the switch between API and transparent mode is determined by the state of the 18<sup>th</sup> pin.

Following are the two configurations selectable by this hardware pin when MODE selection API set as 0.

Pin low: Transparent mode

Pin high: API mode

When the MODE selection API is set to 1, only the API Window is available. The state of the MODE input pin (Pin No. 18) is not taken into consideration.

| Command Structure |               |             |             |            |      |                                                           |
|-------------------|---------------|-------------|-------------|------------|------|-----------------------------------------------------------|
| Start Byte        | Packet Length | Packet Type | Read/Write  | Command ID |      | Value (Hex-1 Byte)                                        |
| 0x2B              | 0x06          | 0x02        | 0x01: Write | 0x00       | 0x16 | 0x00: Pin selection mode<br>0x01: API Mode only (Default) |

| Response Structure |               |             |            |      |                                                                  |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 |
| 0x2B               | 0x05          | 0x82        | 0x00       | 0x16 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID |

Read operation:

| Command Structure |               |             |            |            |      |
|-------------------|---------------|-------------|------------|------------|------|
| Start Byte        | Packet Length | Packet Type | Read/Write | Command ID |      |
| 0x2B              | 0x05          | 0x02        | 0x00: Read | 0x00       | 0x16 |

| Response Structure |               |             |            |      |                                                                  |                                                           |
|--------------------|---------------|-------------|------------|------|------------------------------------------------------------------|-----------------------------------------------------------|
| Start Byte         | Packet Length | Packet Type | Command ID |      | Execution Status                                                 | Value (HEX - 1 Byte)                                      |
| 0x2B               | 0x06          | 0x82        | 0x00       | 0x16 | 0x00: Success<br>0x01: Invalid value<br>0x02: Invalid Command ID | 0x00: Pin selection mode<br>0x01: API Mode only (Default) |

<\*> A flash write and module restart is required for this API write operation to take effect

Note: The module must be in router mode in Transparent mode.